

Dr. Shubham Dadhich

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Motivated | Innovative | Dedicated

I have over 6+ years of professional experience in the field of engineering and research, with expertise in ICT, data analysis, organic thin-film transistors (OTFTs), and semiconductor device modeling. Throughout my career, I have contributed to national-level conferences, participated in advanced research, and developed innovative solutions across various technical domains. My work has not only involved hands-on experience with pioneering technologies but also included mentoring students, preparing academic documentation, and managing large-scale data for educational and research purposes.



EXPERIENCE

20.01.2021 – TILL

ENGINEER GR. II, RIEA, NATIONAL COUNCIL OF EDUCATIONAL RESEARCH & TRAINING (NCERT), GOVT OF INDIA.

I have been responsible for ensuring the smooth operation of ICT section for conferences, workshops, and seminars. I have served as a technical support committee member in various national-level conferences. My experience includes planning and operating networks, overseeing internet usage, optimising load distribution, and managing Google Workspace. Furthermore, **I have earned credits for over 100 Educational Videos published on national platforms such as PM-eVidya TV channels, SWAYAM, and DIKSHA.** Additionally, I have managed large datasets of images and videos with a 100 TB NAS system. I have also contributed as a data analyst for ITEP NCET/CEE admissions and managed data of all admitted previous and current session students using Python. I also prepared comprehensive documents for NAAC accreditation, specifically for Criteria II (Teaching-Learning and Evaluation) and Criteria IV (Infrastructure and Learning Resources).

01.01.2019 – 20.01.2021

JRF AND RESEARCH SCHOLAR, POORNIMA UNIVERSITY, JAIPUR, INDIA

I have conducted advanced research in organic thin-film transistors (OTFTs), focusing on p-type organic semiconductors. My work includes performing TCAD simulations and compact modeling to optimise OTFT device performance. I have authored and co-authored multiple research publications in reputed journals and conferences. Additionally, I have collaborated on projects involving device fabrication, characterisation, and experimental data analysis. I also have teaching experience in B.Tech and M.Tech programs, delivering lectures and laboratory sessions for courses such as Electronic Devices and Circuits (EDC), Microcontrollers and Embedded Systems, Optical Fiber Communication, VLSI Design and Technology, and Digital Image Processing. I have also mentored undergraduate students in semiconductor device modeling and experimental techniques.



EDUCATION

DECEMBER 2024

PhD + M.Tech, POORNIMA UNIVERSITY, JAIPUR

I have completed my PhD in Electrical and Electronics Engineering (EEE) with a thesis titled ***"Numerical Simulation and Compact Modeling of Organic Thin Film Transistors (OTFTs)"***. My research focuses on OTFTs, particularly on p-type organic semiconductors, numerical simulations, and device optimization. I have worked as a Junior Research Fellow (JRF) under the grant **SERB/ECR/2017/000179**, where I contributed to project objectives through advanced simulations, modeling, and experimental data analysis

JULY 2014

B.TECH (ECE), RAJASTHAN TECHNICAL UNIVERSITY, KOTA

Developed a strong foundation in the principles of electronics and communication, with a focus on embedded systems, digital electronics, and communication technologies. As a minor project, I developed a pocket-sized mobile transmission detector capable of sensing an active mobile phone from a distance of up to 1.5 meters. This device is particularly useful for preventing the use of mobile phones in examination halls. As a final project, I designed and implemented a robot based on the **ATmega 8** microcontroller, which was manually controlled through a wired remote interface and featured an integrated camera that streamed live video to a computer. The robot is capable of directional movement and is optimised for navigating flat surfaces efficiently.



AWARDS

Best research paper award at international IEEE conference on:

S. Dadhich, G. Mathur, and A. D. D. Dwivedi, "Mobile Charge Carrier Based Modeling of 4H-21 DNNT and Structure Analysis of OTFT," in *2021 IEEE 2nd International Conference on Technology, Engineering, Management for Societal impact using Marketing, Entrepreneurship and Talent (TEMSMET)*, Pune, India: IEEE, Dec. 2021, pp. 1–5. doi: 10.1109/TEMSMET53515.2021.9768714. (Scopus Indexed)



JOURNAL REVIEWER EXPERIENCE

Served as a reviewer for *IEEE Electron Device Letters*, evaluating manuscripts in the field of organic semiconductor devices and electronic.



SKILL

Simulation and CAD tools: Silvaco TCAD and EDA tools, UTMOST-IV, SmartSpice, H-Spice, Synopsis Sentaurus TCAD.

MATLAB, MATHEMATICA, and Multisim.

Hardware description languages Verilog/ VHDL

Python, C, C++

Embedded System: AVR family (ATmega8/16), PIC middle Range, ARM V7, Arduino, EPS8266/32.

Knowledge and implementation experience of many important sensors: LM35, NTC, IR, Ultrasonic.



PUBLICATION

- [1] Dadhich, S., Dwivedi, A. D. D., & Singh, A. K. (2021). Fabrication, characterization, numerical simulation and compact modeling of P3HT based organic thin film transistors. *Journal of Semiconductors*, 42(7), 074102. doi:10.1088/1674-4926/42/7/074102 [Q1]
- [2] S. Dadhich, G. Mathur, and A. D. D. Dwivedi, "Numerical Simulation, Electrostatic and Physical Compact Modeling of C8-BTBT-C8 Organic Thin Film Transistor," *Inter. Journal of Nanoelectronics and Materials*, vol. 17, no. 2, pp. 2232–1535, Apr. 2024. [Q3]
- [3] Shubham Dadhich, Vivek Upadhyaya, Garima Mathur, Fabrication, Numerical Simulation and Compact Modeling of Ph-BTBT-C10 Organic Thin Film Transistor. *Journal of Polymer and Composites*, (2024) vol. 12, no. 5, pp. 1-18, Aug. 2024 <https://journals.stmjournals.com/jopc/article=2024/view=156433/> (WoS)
- [4] Shubham Dadhich, Vivek Upadhyay, Garima Mathur and Arun Dev Dhar Dwivedi, Fabrication, Compact & Device Modeling of 4H–21DNTT Organic Thin Film Transistor *Engineering Research Express*, online: <https://iopscience.iop.org/article/10.1088/2631-8695/ad8721/meta> [Q3]

As Co-Authors

- [1] A. D. D. Dwivedi, S. K. Jain, R. D. Dwivedi, and S. Dadhich, "Numerical simulation and compact modeling of low voltage pentacene based OTFTs," *Journal of Science: Advanced Materials and Devices*, vol. 4, no. 4, pp. 561–567, Dec. 2019, doi: 10.1016/j.jsamd.2019.10.006. [Q1]
- [2] A. Dev Dhar Dwivedi, S. Kumar Jain, R. Dhar Dwivedi, and S. Dadhich, "Numerical Simulation and Compact Modeling of Thin Film Transistors for Future Flexible Electronics," in *Hybrid Nanomaterials - Flexible Electronics Materials*, R. Vargas-Bernal, P. He, and S. Zhang, Eds., IntechOpen, 2020. doi: 10.5772/intechopen.90301.



CONFERENCES ATTENDED AND PRESENTATIONS

- [1] S. Dadhich, A. D. D. Dwivedi, and G. Mathur, "Numerical Simulation and Analytical Modelling of C8-BTBT-C8 Organic Transistor and Analysis of Semiconductor Thickness," in *Intelligent Computing Techniques for Smart Energy Systems*, vol. 862, A. Tripathi, A. Soni, A. Shrivastava, A. Swarnkar, and J. Sahariya, Eds., in Lecture Notes in Electrical Engineering, vol. 862, Singapore: Springer Nature Singapore, 2022, pp. 669–680. doi: 10.1007/978-981-19-0252-9_60. (Scopus Indexed)
- [2] S. Dadhich, A. D. D. Dwivedi, and G. Mathur, "Surface Charge Based Modeling of TIPS-Pentacene TFT," in *2021 8th International Conference on Smart Computing and Communications (ICSCC)*, Kochi, Kerala, India: IEEE, Jul. 2021, pp. 303–307. doi: 10.1109/ICSCC51209.2021.9528257. (Scopus Indexed)



PROFESSIONAL AFFILIATIONS AND MEMBERSHIPS

- IEEE Membership since 2019.
- IEEE Electron Devices Society Membership.
- IEEE Council on Electronic Design Automation.



REFERENCES

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RIE Ajmer, National Council of Educational
Research and Training (NCERT), GOI, India.
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